

MATH-1014 T10

Tutorial7: Mixed Integration Strategies

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March 23, 2026

Comprehensive Problem

Part III

(a) & (b)

([25 points]) Let $I_n = \int_0^{\frac{\pi}{3}} \tan^n \theta \sec^3 \theta d\theta$, where $n = 0, 1, 2, \dots$ is an integer.

(a) By writing $\tan^n \theta \sec^3 \theta = \tan^{n-1} \theta \cdot (\tan \theta \sec^3 \theta)$ and using integration by parts, show that for $n \geq 2$,

$$I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2} I_{n-2}$$

[10 pts]

(b) Evaluate I_0 .

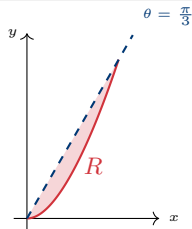
[4 pts]

(Hint: You may need the integral of $\sec \theta$.)

Comprehensive Problem

(c) & (d)

(c) The region R in the first quadrant is enclosed by the polar curve $r = \sqrt{2 \tan^2 \theta \sec^3 \theta}$, the ray $\theta = \frac{\pi}{3}$, and the polar axis ($\theta = 0$). Evaluate the exact area of R . [6 pts]



Solution to (a) - Part 1

Show:
$$I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2} I_{n-2}$$

Let $u = \tan^{n-1} \theta \implies du = (n-1) \tan^{n-2} \theta \sec^2 \theta d\theta$.

Let $dv = \tan \theta \sec^3 \theta d\theta \implies v = \frac{1}{3} \sec^3 \theta$.

Apply Integration by Parts ($\int u dv = uv - \int v du$):

$$I_n = \left[\frac{1}{3} \tan^{n-1} \theta \sec^3 \theta \right]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \frac{1}{3} \sec^3 \theta \cdot (n-1) \tan^{n-2} \theta \sec^2 \theta d\theta$$

1. Evaluate the boundary term: At $\theta = \frac{\pi}{3}$, $\tan(\frac{\pi}{3}) = \sqrt{3}$ and $\sec(\frac{\pi}{3}) = 2$.
At $\theta = 0$, $\tan(0) = 0$ (since $n \geq 2$).

$$\text{Boundary} = \frac{1}{3} (\sqrt{3})^{n-1} (2)^3 - 0 = \frac{8}{3} (\sqrt{3})^{n-1}$$

Solution to (a) - Part 2

2. Simplify the integral term: Replace $\sec^2 \theta$ with $1 + \tan^2 \theta$:

$$I_n = \frac{8}{3}(\sqrt{3})^{n-1} - \frac{n-1}{3} \int_0^{\frac{\pi}{3}} \tan^{n-2} \theta \sec^3 \theta (1 + \tan^2 \theta) d\theta$$

$$I_n = \frac{8}{3}(\sqrt{3})^{n-1} - \frac{n-1}{3} (I_{n-2} + I_n)$$

Multiply the entire equation by 3 to clear denominators:

$$3I_n = 8(\sqrt{3})^{n-1} - (n-1)I_{n-2} - (n-1)I_n$$

$$(3+n-1)I_n = 8(\sqrt{3})^{n-1} - (n-1)I_{n-2}$$

$$(n+2)I_n = 8(\sqrt{3})^{n-1} - (n-1)I_{n-2}$$

Divide by $(n+2)$ to obtain the final formula:

$$I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2} I_{n-2}$$

Solution to (b) & (c)

(b) Evaluate I_0 :

$$\begin{aligned} I_0 &= \int_0^{\frac{\pi}{3}} \sec^3 \theta \, d\theta = \frac{1}{2} \left[\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta| \right]_0^{\frac{\pi}{3}} \\ &= \frac{1}{2} \left(2\sqrt{3} + \ln(2 + \sqrt{3}) \right) - 0 = \boxed{\sqrt{3} + \frac{1}{2} \ln(2 + \sqrt{3})} \end{aligned}$$

(c) Evaluate Polar Area:

$$A = \frac{1}{2} \int_0^{\frac{\pi}{3}} r^2 \, d\theta = \frac{1}{2} \int_0^{\frac{\pi}{3}} 2 \tan^2 \theta \sec^3 \theta \, d\theta = \int_0^{\frac{\pi}{3}} \tan^2 \theta \sec^3 \theta \, d\theta$$

This is exactly I_2 . Apply the reduction formula from (a) with $n = 2$:

$$\begin{aligned} I_2 &= \frac{8(\sqrt{3})^1}{4} - \frac{1}{4} I_0 = 2\sqrt{3} - \frac{1}{4} \left(\sqrt{3} + \frac{1}{2} \ln(2 + \sqrt{3}) \right) \\ &= \boxed{\frac{7\sqrt{3}}{4} - \frac{1}{8} \ln(2 + \sqrt{3})} \end{aligned}$$

The "Mixed Methods" Approach

In exams, you rarely use just one technique. Be prepared to chain them together based on the 4-Step Strategy:

① Substitution → Partial Fractions:

Often triggered by e^x , $\ln x$, or trig functions hiding inside rational structures.

② Rationalizing Substitution → Long Division:

When dealing with multiple radicals like $\sqrt{x}$ and $\sqrt[n]{x}$, let $u = \sqrt[n]{x}$ where n is the **LCM** (Least Common Multiple) of the indices. This transforms it into an improper rational function.

③ Algebraic Manipulation → Standard Forms:

Multiplying by the conjugate (e.g., $\frac{1+\cos x}{1+\cos x}$) to exploit identities like

$$1 - \cos^2 x = \sin^2 x.$$

④ Substitution → Integration by Parts (IBP):

Functions like $e^{\sqrt{x}}$ or $\sin(\ln x)$ require substitution first to reveal the IBP structure.

Problem 1: The LCM Rationalizing Substitution

Exercise 1

Evaluate the integral:

$$\int \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx$$

Strategic Hint:

You have $x^{1/2}$ and $x^{1/3}$. To eliminate BOTH fractional powers, let $x = u^n$ where n is the Least Common Multiple of 2 and 3.

After substitution, check if you need polynomial long division!

Solution to Problem 1

1. Substitution:

Let $x = u^6$. Then $dx = 6u^5 du$. Also, $\sqrt{x} = u^3$ and $\sqrt[3]{x} = u^2$.

$$\int \frac{1}{u^3 + u^2} \cdot 6u^5 du = \int \frac{6u^5}{u^2(u+1)} du = 6 \int \frac{u^3}{u+1} du$$

2. Long Division:

Since degree 3 $\geq$ degree 1, we divide u^3 by $(u+1)$:

$$\frac{u^3}{u+1} = \frac{(u^3 + 1) - 1}{u+1} = (u^2 - u + 1) - \frac{1}{u+1}$$

3. Integration & Back-Substitution:

$$\begin{aligned} 6 \int \left(u^2 - u + 1 - \frac{1}{u+1} \right) du &= 6 \left(\frac{u^3}{3} - \frac{u^2}{2} + u - \ln|u+1| \right) + C \\ &= \boxed{2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6\ln|\sqrt[6]{x} + 1| + C} \end{aligned}$$

Problem 2: Hidden Trig Substitution & Case III

Exercise 2

Evaluate the integral:

$$\int \frac{\cos x}{\sin^3 x + \sin x} dx$$

Strategic Hint:

Notice the $\cos x dx$ in the numerator? That's the derivative of $\sin x$.

After substitution, you will face a denominator with an **irreducible quadratic factor** (Case III in Partial Fractions).

Solution to Problem 2

1. Substitution:

Let $u = \sin x$, then $du = \cos x dx$.

$$\int \frac{1}{u^3 + u} du = \int \frac{1}{u(u^2 + 1)} du$$

2. Partial Fractions (Case III):

$$\frac{1}{u(u^2 + 1)} = \frac{A}{u} + \frac{Bu + C}{u^2 + 1}$$

Multiply by $u(u^2 + 1)$: $1 = A(u^2 + 1) + (Bu + C)u = (A + B)u^2 + Cu + A$.

Comparing coefficients: $A = 1, C = 0, A + B = 0 \implies B = -1$.

$$\frac{1}{u(u^2 + 1)} = \frac{1}{u} - \frac{u}{u^2 + 1}$$

3. Integrate:

$$\int \left(\frac{1}{u} - \frac{u}{u^2 + 1} \right) du = \ln |u| - \frac{1}{2} \ln(u^2 + 1) + C$$

$$= \ln |\sin x| - \frac{1}{2} \ln(\sin^2 x + 1) + C$$

Problem 3: Factoring by Grouping & Completing the Square

Exercise 3

Evaluate the integral:

$$\int \frac{3x^2 - 2x + 5}{x^3 - x^2 + 2x - 2} dx$$

Strategic Hint:

Factor the denominator by grouping terms: $x^2(x - 1) + 2(x - 1)$.

Once decomposed into partial fractions, you will need to split the numerator of the quadratic term to use both $\ln()$ and $\arctan()$.

Solution to Problem 3

1. Factor & Partial Fractions:

Denominator: $x^3 - x^2 + 2x - 2 = (x - 1)(x^2 + 2)$.

$$\frac{3x^2 - 2x + 5}{(x - 1)(x^2 + 2)} = \frac{A}{x - 1} + \frac{Bx + C}{x^2 + 2}$$

$$3x^2 - 2x + 5 = A(x^2 + 2) + (Bx + C)(x - 1).$$

$$\text{Let } x = 1 \implies 6 = 3A \implies \mathbf{A = 2}.$$

$$x^2 \text{ terms: } 3 = A + B \implies \mathbf{B = 1}.$$

$$\text{Constant terms: } 5 = 2A - C \implies 5 = 4 - C \implies \mathbf{C = -1}.$$

2. Integrate:

$$\begin{aligned} \int \left(\frac{2}{x-1} + \frac{x-1}{x^2+2} \right) dx &= \int \frac{2}{x-1} dx + \int \frac{x}{x^2+2} dx - \int \frac{1}{x^2+2} dx \\ &= \boxed{2 \ln |x-1| + \frac{1}{2} \ln(x^2+2) - \frac{1}{\sqrt{2}} \arctan \left(\frac{x}{\sqrt{2}} \right) + C} \end{aligned}$$

Problem 4: Algebraic Manipulation (The Conjugate Trick)

Exercise 4

Evaluate the integral:

$$\int \sqrt{\frac{1-x}{1+x}} dx$$

Strategic Hint:

This looks like a trigonometric substitution candidate, but there's a faster way. Multiply the numerator and denominator inside the square root by $(1-x)$ to create a perfect square on top and a difference of squares on the bottom.

Solution to Problem 4

1. Algebraic Manipulation:

Multiply by $\sqrt{\frac{1-x}{1+x}}$:

$$\int \sqrt{\frac{1-x}{1+x}} \cdot \sqrt{\frac{1-x}{1-x}} dx = \int \frac{1-x}{\sqrt{1-x^2}} dx$$

2. Split into Two Standard Integrals:

$$\int \frac{1-x}{\sqrt{1-x^2}} dx = \int \frac{1}{\sqrt{1-x^2}} dx - \int \frac{x}{\sqrt{1-x^2}} dx$$

3. Integrate: The first integral is standard arcsine. For the second, let $u = 1 - x^2$, $du = -2x dx$.

$$\begin{aligned} \int \frac{1}{\sqrt{1-x^2}} dx - \left(-\frac{1}{2} \int \frac{1}{\sqrt{u}} du \right) &= \arcsin x + \sqrt{u} + C \\ &= \boxed{\arcsin x + \sqrt{1-x^2} + C} \end{aligned}$$

Problem 5: Substitution into Integration by Parts

Exercise 5

Evaluate the integral:

$$\int e^{\sqrt{x}} dx$$

Strategic Hint:

Whenever you see a non-linear function trapped inside an exponential or trigonometric function, your first move should be substitution to "free" it.

Let $u = \sqrt{x}$. After substitution, you will be left with a product of an algebraic and an exponential function. What method handles products?

Solution to Problem 5

1. Substitution:

Let $u = \sqrt{x}$. Then $x = u^2$ and $dx = 2u du$.

$$\int e^{\sqrt{x}} dx = \int e^u \cdot 2u du = 2 \int ue^u du$$

2. Integration by Parts:

Use $\int w dv = wv - \int v dw$.

Let $w = u \implies dw = du$.

Let $dv = e^u du \implies v = e^u$.

$$\begin{aligned} 2 \int ue^u du &= 2 \left(ue^u - \int e^u du \right) \\ &= 2(ue^u - e^u) + C \end{aligned}$$

3. Back-Substitute:

$$\boxed{2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C}$$